

Références : -Transport phenomena, 2nd edition.
-gch200.pheno.ca/ → Chap2.pdf

Outils graphiques pour aider à visualiser les différents termes des bilans des forces sur un élément de fluide.

Les équations suivantes résument toutes les forces qui s'exercent sur un fluide en mouvement en 3 dimensions, les pages suivantes permettent de visualiser les surfaces du parallélépipède sur lequel le bilan est effectué

Table 1.7-2 Summary of Notation for Momentum Fluxes

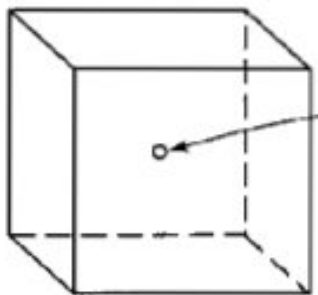
Symbol	Meaning	Reference
$\rho \mathbf{v} \mathbf{v}$	Convective momentum-flux tensor	Table 1.7-1
$\boldsymbol{\tau}$	Viscous momentum-flux tensor ^a	Table 1.2-1
$\boldsymbol{\pi} = p\boldsymbol{\delta} + \boldsymbol{\tau}$	Molecular momentum-flux tensor ^b	Table 1.2-1
$\boldsymbol{\phi} = \boldsymbol{\pi} + \rho \mathbf{v} \mathbf{v}$	Combined momentum-flux tensor	Eq. 1.7-2

Table 1.2-1 Summary of the Components of the Molecular Stress Tensor (or Molecular Momentum-Flux Tensor)^a

Direction normal to the shaded face	Vector force per unit area on the shaded face (momentum flux through shaded face)	Components of the forces (per unit area) acting on the shaded face (components of the momentum flux through the shaded face)		
		x-component	y-component	z-component
x	$\boldsymbol{\pi}_x = p\boldsymbol{\delta}_x + \boldsymbol{\tau}_x$	$\pi_{xx} = p + \tau_{xx}$	$\pi_{xy} = \tau_{xy}$	$\pi_{xz} = \tau_{xz}$
y	$\boldsymbol{\pi}_y = p\boldsymbol{\delta}_y + \boldsymbol{\tau}_y$	$\pi_{yx} = \tau_{yx}$	$\pi_{yy} = p + \tau_{yy}$	$\pi_{yz} = \tau_{yz}$
z	$\boldsymbol{\pi}_z = p\boldsymbol{\delta}_z + \boldsymbol{\tau}_z$	$\pi_{zx} = \tau_{zx}$	$\pi_{zy} = \tau_{zy}$	$\pi_{zz} = p + \tau_{zz}$

$$\Phi = \pi + \rho \mathbf{v} \mathbf{v} = p \delta + \tau + \rho \mathbf{v} \mathbf{v}$$

$$\phi_{xx} = \pi_{xx} + \rho v_x v_x = p + \tau_{xx} + \rho v_x v_x$$

$$\phi_{xy} = \pi_{xy} + \rho v_x v_y = \tau_{xy} + \rho v_x v_y$$


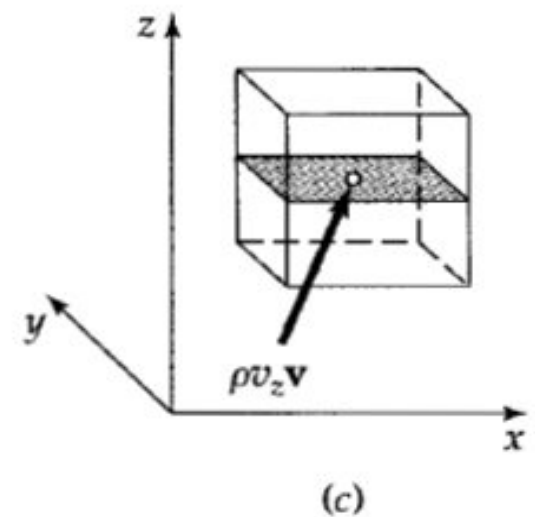
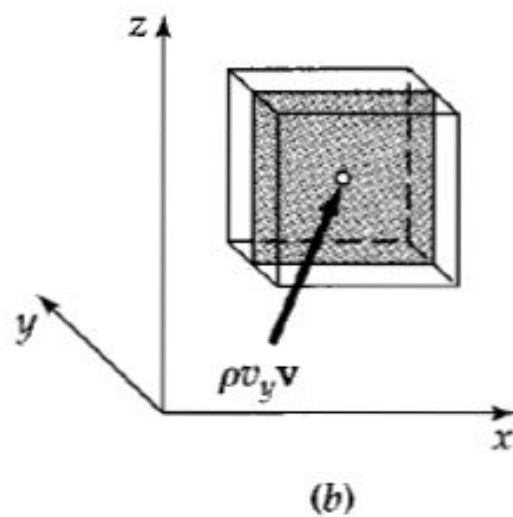
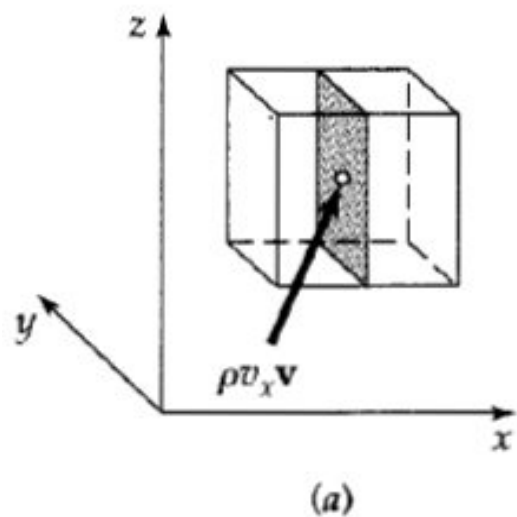


Fig. 1.7-1 The convective momentum fluxes through planes of unit area perpendicular to the coordinate directions.

Rappels importants sur le chapitre 1

Détail des forces agissant sur les éléments de volume de fluide

Composantes
convectives

$$\rho v_x v_x + \tau_{xx} + P$$

$$\rho v_y v_x + \tau_{yx}$$

$$\rho v_z v_x + \tau_{zx}$$

$$\rho v_x v_y + \tau_{xy}$$

$$\rho v_y v_y + \tau_{yy} + P$$

$$\rho v_z v_y + \tau_{zy}$$

$$\rho v_x v_z + \tau_{xz}$$

$$\rho v_y v_z + \tau_{yz}$$

$$\rho v_z v_z + \tau_{zz} + P$$

Composantes
visqueuses

$$\rho v_x v_x + \tau_{xx} + P$$

$$\rho v_x v_y + \tau_{xy}$$

$$\rho v_x v_z + \tau_{xz}$$

$$\rho v_y v_x + \tau_{yx}$$

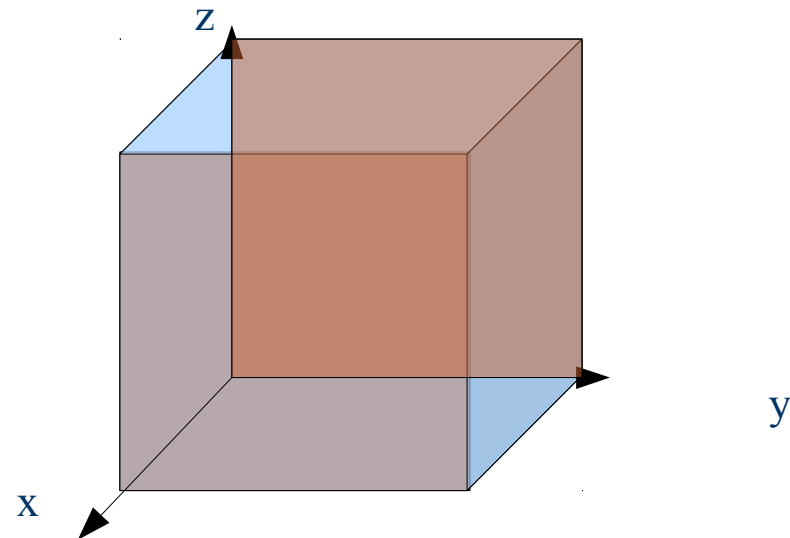
$$\rho v_y v_y + \tau_{yy} + P$$

$$\rho v_y v_z + \tau_{yz}$$

$$\rho v_z v_x + \tau_{zx}$$

$$\rho v_z v_y + \tau_{zy}$$

$$\rho v_z v_z + \tau_{zz} + P$$



$$\rho v_x v_x + \tau_{xx} + P$$

$$\rho v_x v_y + \tau_{xy}$$

$$\rho v_x v_z + \tau_{xz}$$

$$\rho v_y v_x + \tau_{yx}$$

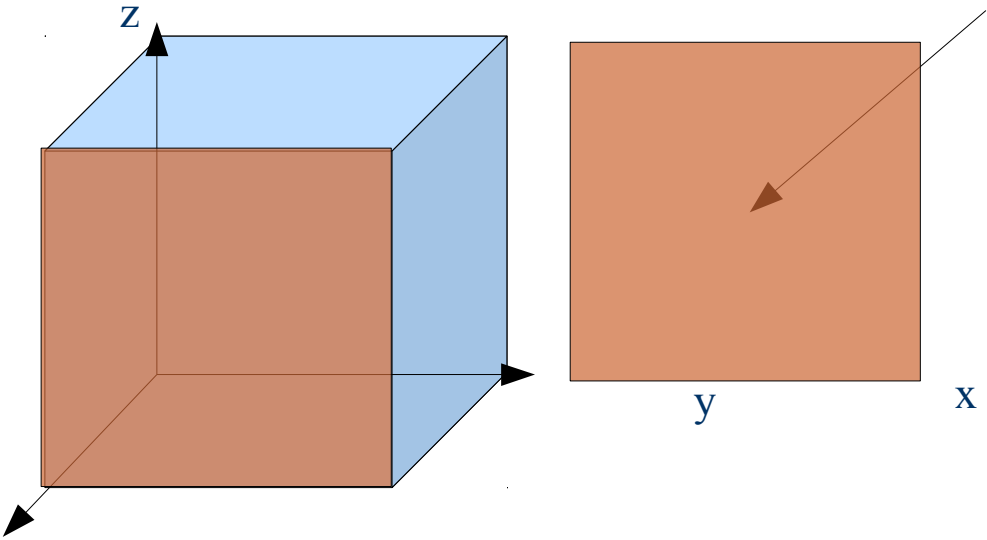
$$\rho v_y v_y + \tau_{yy} + P$$

$$\rho v_y v_z + \tau_{yz}$$

$$\rho v_z v_x + \tau_{zx}$$

$$\rho v_z v_y + \tau_{zy}$$

$$\rho v_z v_z + \tau_{zz} + P$$



$$\rho v_x v_x + \tau_{xx} + P$$

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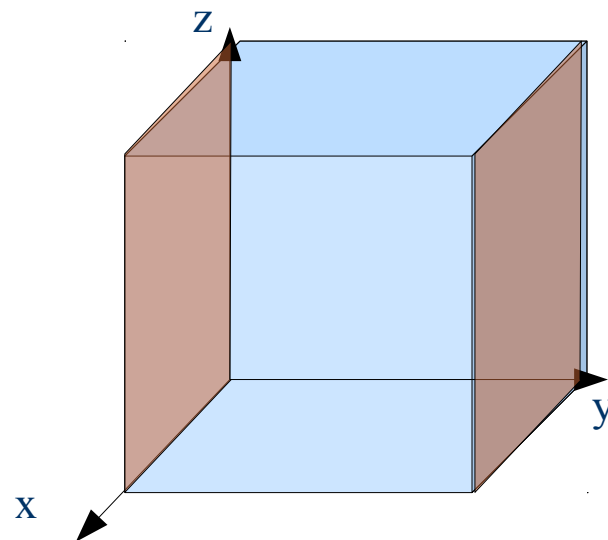
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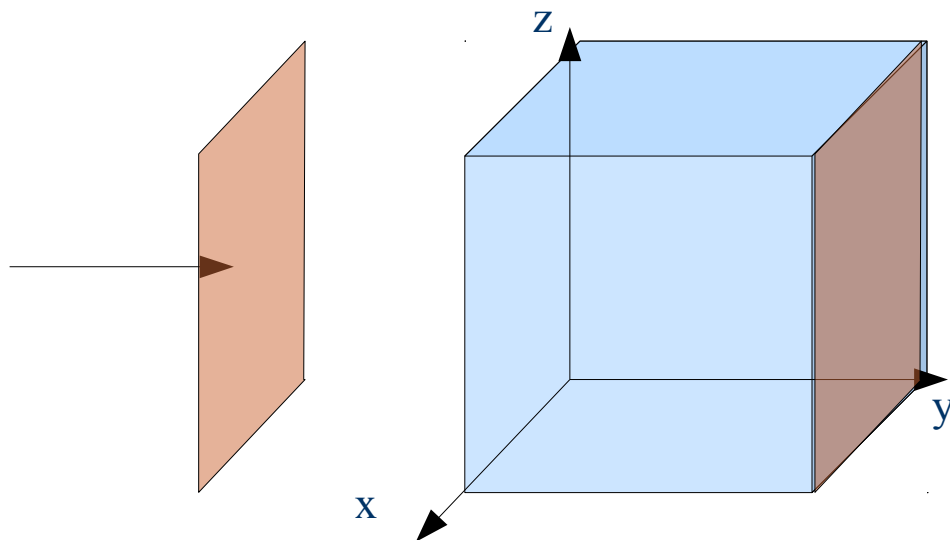
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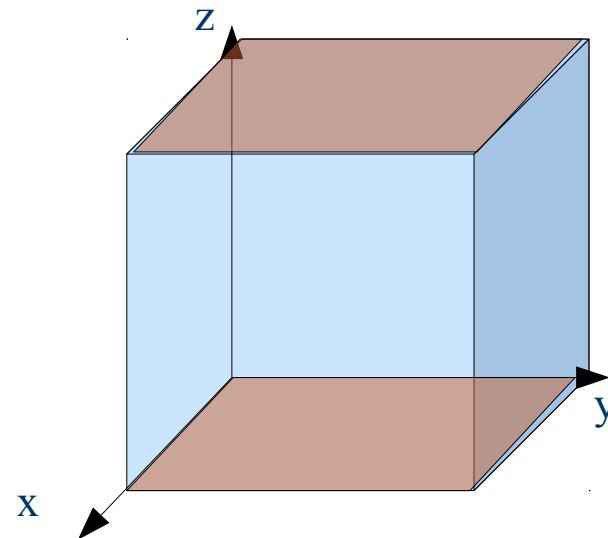
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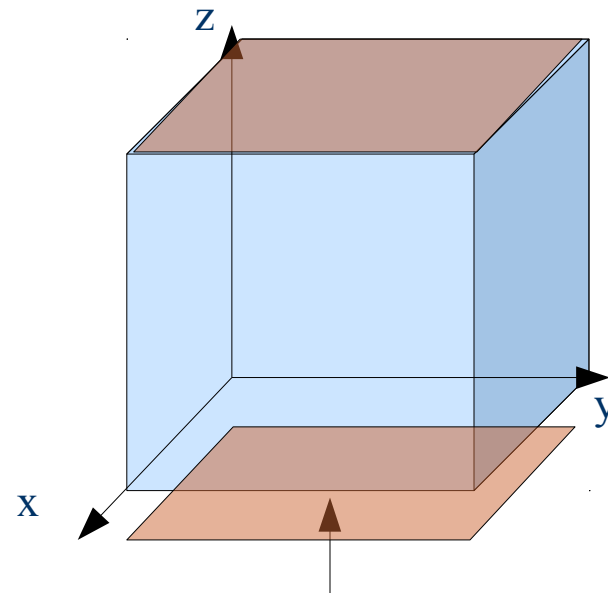
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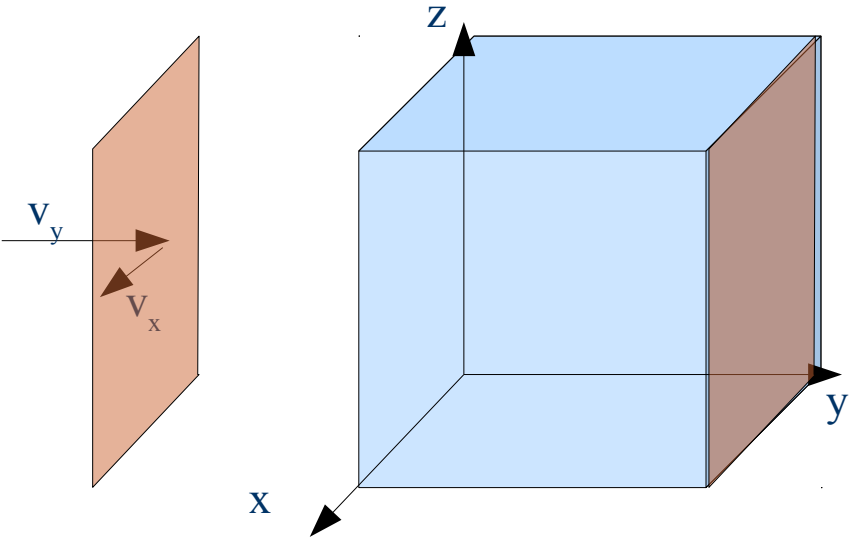
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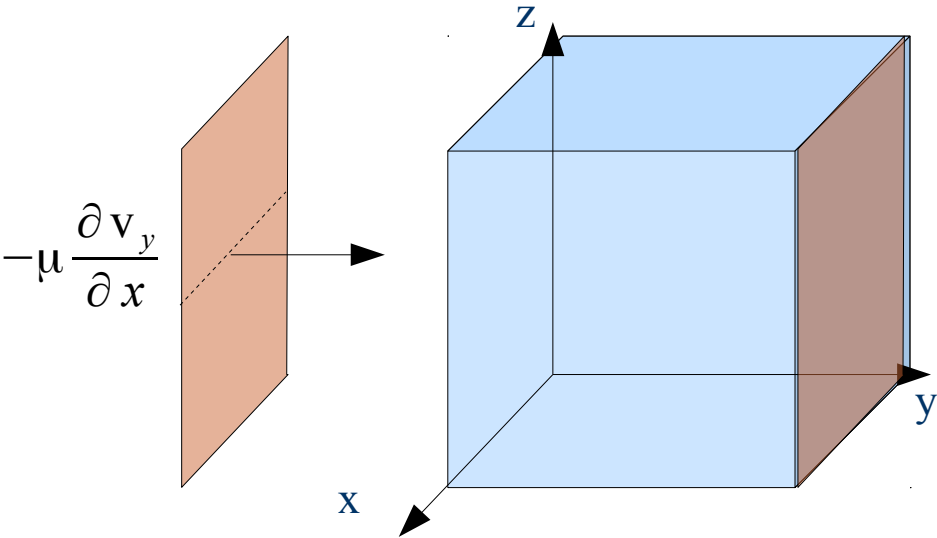
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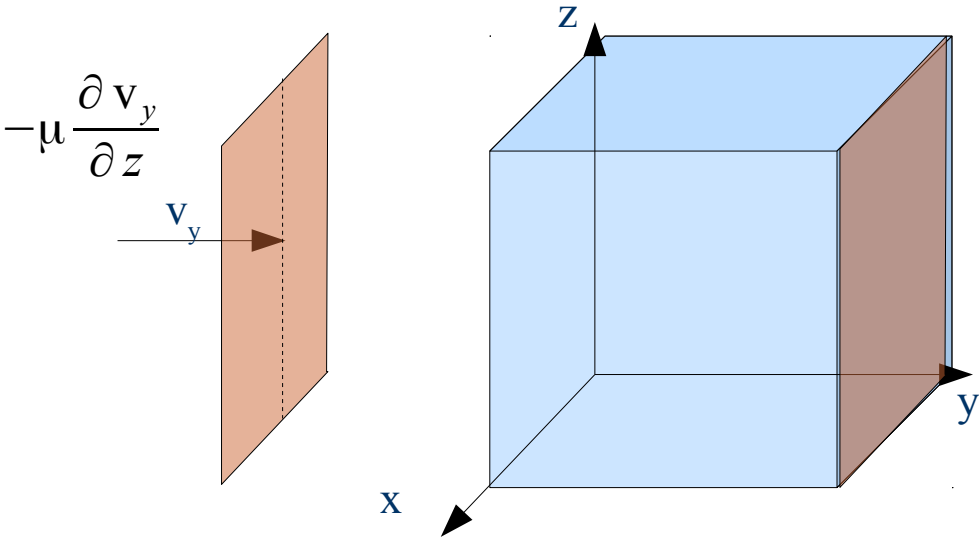
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